

A Novel Decision Support System for Optimizing Aircraft Maintenance Check Schedule and Task Allocation

Abstract

Modern aircraft have thousands of parts, systems, and components that need to be recurrently inspected or replaced. To keep the fleet airworthy, maintenance planners have to schedule the maintenance checks for each aircraft and the associated tasks. In practice, these two complex problems are solved following the experience of planners, resulting in sub-efficient solutions. This paper presents the first decision support system (DSS) developed for optimizing both aircraft maintenance check schedule and task allocation. The DSS integrates aircraft maintenance check scheduling, task allocation to each maintenance check, and shift planning in the same framework. The practical relevance of the DSS is illustrated through three test cases. The results show that the DSS can be used not only to optimize maintenance plans but also to study future maintenance policies. The results reveal substantial improvements in all key performance indicators compared with the planning approach followed by a partner airline.

Keywords: Decision Support, Aircraft Maintenance, Task Allocation, Dynamic Programming, Bin Packing

1. Introduction

Aircraft maintenance is a sequence of activities, including overhaul, repair, inspection, or modification of an aircraft or aircraft systems, components, and structures to ensure an aircraft retains an airworthy condition. In the aviation industry, a commercial aircraft must undergo regular maintenance to prevent component and system failures during operations. Many of the aircraft main-

7 tenance activities take place after an aircraft has been operating certain flight
8 hours (FH), flight cycles (FC), or calendar days (DY). The FH, FC, and DY
9 are known as usage parameters to indicate aircraft utilization. The maximum
10 usage parameters allowed in operation are defined as inspection intervals.

11 Modern aircraft have thousands of parts, systems, and components that
12 need to be recurrently inspected, serviced, and replaced. Many airlines adopt a
13 top-down approach to plan aircraft maintenance:

14 • **Step 1 – Maintenance Check Scheduling**

15 First group major maintenance tasks with the same or similar inspection
16 intervals into letter checks: A-, B-¹, C- and D-check, as showed in Table
17 1. Each check type is coupled with an elapsed time (time required for the
18 execution of tasks within letter checks + time estimated for other tasks).
19 Maintenance planners then create a letter check schedule (3–5 years for C-
20 /D-check and 6–12 months for A-check) according to pre-defined elapsed
21 time of each check type. The letter checks are performed in the hangar.

22 • **Step 2 – Maintenance Task Allocation**

23 Although some tasks can quickly be packaged into the letter checks, a large
24 number of other tasks (e.g., more than 70% for an Airbus A320 aircraft)
25 are dephased from the inspection intervals of these checks. It means that
26 they either have to be manually allocated by maintenance operators to
27 different maintenance events based on the suitability of the task to that
28 check and the urgency of performing the task.

29 Despite the rapid expansion of the global air travel industry and the increase
30 of fleet size, the advances in aircraft maintenance planning (AMP) have been
31 struggling to keep up with the times. In practice, AMP involves scheduling
32 maintenance checks to each aircraft, allocating tasks to each check, planning
33 the workforce for each task, inventory optimization, and coordination of main-

¹B-checks are rarely mentioned in practice. The tasks within B-checks are commonly incorporated into successive A-checks.

Table 1: Aircraft letter check and corresponding inspection interval [1].

Check Type	Interval	Maintenance Tasks
A-check	2-3 months	External visual inspection, filter replacement, etc.
B-check	—	Rarely mentioned in practice
C-check	18-24 months	Inspection of the individual systems and components
D-check	6-10 years	Inspection of most structurally significant items

tenance tools. For small airlines, AMP is not so demanding and can be done manually according to the experience of maintenance planners. For large airlines, the AMP problem becomes more complex – maintenance planners have to spend several days or weeks on scheduling maintenance activities because of the lack of efficient tools. Since, on average, 9%—10% of the total cost of airlines goes to aircraft maintenance, which is equivalent to about \$2.5M per aircraft per year [2], the savings derived from efficient AMP can be very substantial.

To facilitate the AMP process, many companies engage in developing AMP systems. For example, Ref. [3] developed one of the first AMP tools to improve maintenance efficiency and reduce associated cost. After that, many companies followed and developed various tools, e.g., Solumina MRO from iBASEt, Airline Suite from C.A.L.M Systems INC., WinAir from AV-Base Systems, and Mainenix from IFS, etc. To our best knowledge, all the available commercial tools focus on managing and tracking the status of the maintenance tasks, providing a valuable computer-aid solution to manual planning. However, none of them has the function of producing an optimized maintenance schedule automatically.

AMP is challenging due to the lack of optimization approaches for planning maintenance checks and associated tasks, even though there are many available computer-aid solutions. Two distinct limitations in the current academic and industrial state of the art can be discerned, as further discussed in Section 2: 1) a lack of decision support system (DSS) to optimize the maintenance check (A-, B-, C- and D-checks) schedule; 2) a lack of DSS for optimizing aircraft maintenance check and task execution in an integrated manner. In the literature, there is no work integrating the two problems in a single optimization framework.

58 In 2015, the AIRMES project was launched by Clean Sky Joint Undertak-
59 ing, a public-private partnership between the European Commission and the
60 European aeronautics industry, to optimize end-to-end maintenance activities
61 within an airline operator’s environment [4]. We developed a DSS during the
62 project to automate the maintenance planning process and provide maintenance
63 check scheduling optimization, optimal task allocation, and shift planning in one
64 comprehensive solution. The contribution of our research is threefold:

- 65 • The DSS integrates aircraft maintenance check scheduling, maintenance
66 task allocation, and work shift planning in the same framework. In prac-
67 tice, these processes are solved using different tools, while in the literature,
68 these are seen as three different problems handled separately.
- 69 • We demonstrate that the DSS can improve aircraft utilization and re-
70 duce maintenance costs, compared with the current practice of airlines. It
71 reduces the time needed for AMP from days or hours to 20–30 minutes.
- 72 • We also present the usefulness of the DSS in helping airlines evaluate
73 different aircraft maintenance strategies before implementation.

74 This paper presents the architecture of the resulting DSS and the correspond-
75 ing optimization modules for maintenance check schedule, task allocation, and
76 shift planning. We also discuss the applicability of the DSS by presenting the
77 results from a case study with one major European airline and several industry
78 partners. The case study validates the utility of the DSS for both maintenance
79 planning optimization and future scenario analysis.

80 The outline of this paper is as follows: Section 2 gives an overview of the
81 relevant literature on the aircraft maintenance domain. The DSS architecture
82 is presented in Section 3, aircraft maintenance check scheduling optimization,
83 optimal task allocation, and shift planning as well as their corresponding al-
84 gorithms. Section 4 describes the demonstration exercise with data from the
85 partner airline. The last section summarizes the research with concluding re-
86 marks and gives an outlook on future work.

87 2. Related Work

88 The aviation industry is extremely competitive in Europe. The average net
89 profit of airlines usually represents only up to 4%–5% of revenues and about
90 9%–10% of the total cost goes to aircraft maintenance [2]. Efficient AMP is
91 one useful way of reducing maintenance costs. The benefit of efficient AMP
92 is two-fold: on the one hand, the increased aircraft availability indicates that
93 there will be more aircraft available for commercial operations, and eventually,
94 generating more revenues; on the other hand, it decreases the number of air-
95 craft maintenance inspections, and therefore, reduces the maintenance operation
96 costs in the long term. This section reviews the previous research on AMP from
97 long-term planning (3–5 years) to short-term planning (several days to weeks).

98 2.1. Long-Term Aircraft Maintenance Planning

99 Long-term AMP aims to generate an aircraft heavy maintenance schedule
100 (C- and D-checks) before determining the tasks within each check, also known as
101 aircraft maintenance check scheduling (AMCS). It is indispensable since C-check
102 has an interval of 18–24 months, and D-check is usually scheduled once every
103 6 years; airlines need a C- and D-check schedule to further plan the A- and B-
104 checks and the associated tasks for all the (A-, B-, C-, and D-) checks. In 1977,
105 Air Canada developed one of the first DSSs for the long-term AMCS, called
106 AMOS [3]. AMOS was considered a computer-aid manual planning approach
107 since the developers did not see the value of finding an optimal solution that
108 could rapidly become obsolete due to uncertainty. It helped Air Canada reduce
109 the time for planning a 5-year C-check schedule for its fleet from 3 weeks to a
110 few hours. Besides, Ref. [3] defined the long-term (3–5 years) planning, and it
111 is the only available reference of the long-term AMP category before 2020.

112 Following this research direction, Ref. [5] proposed a dynamic programming
113 (DP) based methodology for long-term AMCS within the AIRMES project in
114 2020, adopting the assumptions, problem formulation presented in [3]. It aimed
115 to optimize the aircraft maintenance check schedule for the future 3–5 years.

116 This work is the first step towards building an integrated AMP framework, fo-
117 cusing on long-term AMP. The DP-based methodology generates an optimized
118 4-year schedule for both light and heavy maintenance within 15 minutes. The
119 optimized maintenance check schedule can be further used to plan the mainte-
120 nance tasks within each check and daily work shift.

121 *2.2. Short-Term Aircraft Maintenance Planning*

122 In contrast to the little available literature about long-term AMCS, there are
123 many studies on short-term AMP in the topics of aircraft maintenance routing,
124 maintenance personnel management, and maintenance task scheduling. The
125 reason is that by optimizing short-term maintenance activities, airlines can see
126 tangible benefits in a few days or weeks.

127 *Aircraft Maintenance Routing*

128 Aircraft maintenance routing (AMR) is to design flight routes for every air-
129 craft to meet the maintenance requirements set by Federal Aviation Adminis-
130 tration (FAA) and individual airline companies. Extensive research works have
131 contributed to AMR through flight schedule design [6, 7], determining routes
132 flown by each aircraft [8–10], fleet assignment (assigning an aircraft model for
133 each flight) [11–13], tail assignment (determining which aircraft should fly which
134 segment) [14–16], or even addressing the aircraft routing in conjunction with
135 crew pairing [17–19]. These studies usually consider aircraft maintenance as an
136 operational requirement but did not plan the maintenance checks or tasks.

137 *Maintenance Personnel Planning*

138 Maintenance personnel planning (MPP) is one of the main research direc-
139 tions of short-term AMP. An effective maintenance workforce supply can reduce
140 operations costs while ensuring aviation safety and punctuality. It has attracted
141 lots of attention from both industry and academia. Early in 1994, KLM Royal
142 Dutch Airline and Erasmus University Rotterdam developed a DSS to smooth
143 the workload of aircraft maintenance personnel by increasing the number of
144 peaks of workloads and reducing the peak length [20]. It helped KLM improved

the utilization of maintenance technicians (the ratio of productivity labor-hours to total available labor-hour). After that, many researchers envisioned the potential benefits and continued the MPP study, such as optimizing the workforce supply [21–23], or minimizing the total labor cost [24, 25]. However, MPP usually assumes that maintenance tasks are given rather than planning the tasks.

Maintenance Task Scheduling

Maintenance task scheduling (MTS) refers to allocating maintenance tasks to time slots so that the tasks can be executed before due dates. It includes task scheduling for aircraft line maintenance (coordinating maintenance tasks to be carried out at the gate during turnaround time and the required maintenance resources), daily hangar maintenance, or work shift. There are some studies addressing the MTS for line maintenance, such as spreading the workload more uniformly across shifts [26], improving aircraft availability and reducing maintenance costs [27], or optimizing both workforce and tasks [28]. MTS for line maintenance planning has an operational nature. It only focuses on optimizing a limited number of maintenance tasks during aircraft turnaround time.

Task scheduling for daily aircraft hangar maintenance can be seen in [29]. According to the authors, optimizing the daily hangar maintenance tasks to be executed 24 hours beforehand also maximizes the availability of fighting jets for the missions of the next day. The authors call attention to the fact that if we want to plan the daily maintenance task for each letter check, we have to look into a planning horizon longer than 24 hours, especially for the C-/D-check. Besides, the daily maintenance task plan bridges the gap between AMCS and associated work shift planning. That is, we can better plan each morning/afternoon/evening shift and prepare the tools and aircraft spare parts if we know the daily maintenance tasks in advance. Hence, Ref. [30] proposed a bin packing approach to determine daily maintenance tasks (for each A-/B-/C-/D-check) given a long-term (3–5 years) maintenance check schedule for AIRMES. As a result, it gives a long-term (3–5 years) plan of maintenance tasks for each day and a heterogeneous fleet of aircraft.

175 2.3. Concluding Remarks for Literature Review

176 To our best knowledge, most of the studies in the AMP domain focus ei-
177 ther on AMR or MPP, assuming that the maintenance tasks are given. There
178 are some studies on MTS, yet most of them focus on line maintenance prob-
179 lems. The long-term and short-term AMP was not yet considered in a single
180 framework, nor was a DSS presented in the literature addressing the AMP. Syn-
181 thesizing the literature review gives rise to two challenges in the AMP domain:

- 182 1. No DSS for aircraft maintenance planning optimization (AMPO) is pre-
183 sented in the academic literature that can generate an optimally integrated
184 maintenance check and task execution plan at the fleet level.
- 185 2. Commercial DSSs addressing the fleet maintenance check level are rela-
186 tively rare. Even so, they do not optimize the maintenance check schedule.

187 In practice, maintenance planners have to spend a significant amount of
188 time and effort scheduling the aircraft letter checks and coordinating associated
189 tasks execution activities. It can happen that with the aid of current DSSs, the
190 maintenance planners still obtain an inefficient plan; this may, in the long-term,
191 result in more letter checks and higher operation costs.

192 The DSS presented in this paper contributes to bridging two main research
193 streams, long-term and short-term AMP, by integrating the AMCS problem and
194 its methodology presented in [5], the MTS problem and the associated algorithm
195 presented in [30], and a shift planning approach into the same framework.

196 3. System Architecture

197 To address the challenges identified in Section 2, we developed a DSS specif-
198 ically for AMP using the programming language Python and for Windows op-
199 erating system. The DSS is a stand-alone software prototype and has already
200 been converted to an executable file. It can be run on any individual PC with-
201 out installation or a license. The DSS consists of three components (layers), a
202 database, a model, and a graphical user interface (GUI):

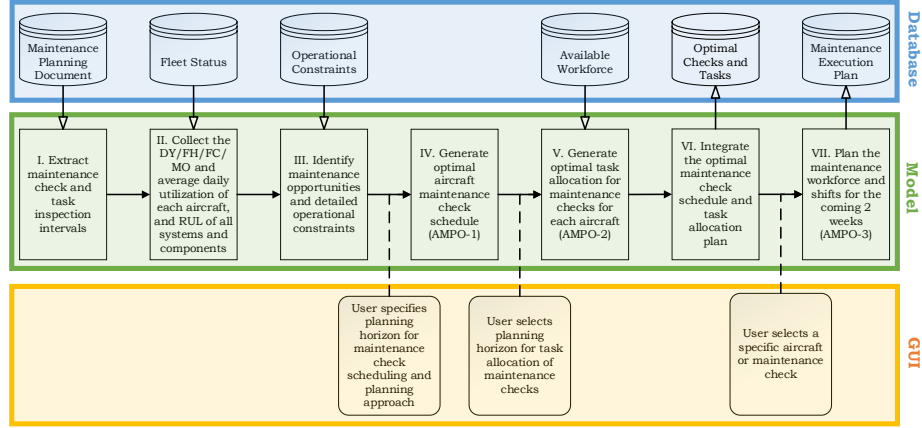


Figure 1: Architecture of the decision support system for AMPO.

- *Database*: Store the input data, including the maintenance planning document (MPD) for aircraft manufacturers, fleet status, operational constraints, and available workforce from airlines.
- *Model*: Clean and process input data, optimize the aircraft maintenance check schedule and maintenance task execution plan.
- *Graphical User Interface (GUI)*: Allow users to interact with the DSS and visualize the planning results and the associated KPIs.

In this section, we present the structure of the DSS layer by layer, as illustrated in Figure 1. We begin with description of database layer (Subsection 3.1) and input, followed by a detailed introduction of the optimization models and algorithm (Subsection 3.2). In subsection 3.3, we outline the GUI of the DSS.

3.1. Database and Input Data

The database stores the input in the format of comma-separated values (CSV) and output in Excel. We classify the input into four categories:

Maintenance Planning Document

The maintenance planning document (MPD) is provided by the aircraft manufacturer. It specifies the maintenance tasks according to the aircraft structure,

Table 2: An example of fleet status with respect to aircraft C-Check on 01/07/2020.

Fleet	Tail	Before	Next	DY	FH	FC	fh/day	fc/day	Phase-In
A320	AC-1	C 12.1	-1	212	2391	963	10.3	4.2	12/01/1998
A319	AC-2	C 10.1	C 11.1	607	6439	2600	9.9	4.1	08/06/1998
A321	AC-3	—	C 1.1	0	0	0	10.1	4.2	01/03/2021

systems, and components, as well as corresponding inspection intervals (as described in Table 1). The MPD gives strict criteria for aircraft maintenance – all letter checks and tasks have to be performed before the corresponding usage parameters reached their maximums (intervals). Any violation of maintenance task execution will prevent the aircraft from flying because of safety concerns.

Fleet Status

We use Table 2 to illustrate the structure input data. The column **Fleet** shows the aircraft type. **Tail No.** indicates the aircraft tail number. **Before** and **Next** represent the previous and next letter checks respectively. **DY**, **FH** and **FC** are the usage parameters of the fleet. **fh/day** and **fc/day** are the average daily utilization of the fleet. **Phase-In** indicates when an aircraft starts in commercial operation. This is relevant information as old aircraft will phase out after a certain number of checks, and meanwhile, airlines have new aircraft in operation. If an aircraft will phase out, we give “-1” to its next A-/C-/D-check, meaning that no more A-/C-/D-check needs to be scheduled. If the phase-in date of an aircraft is later than the current date, this aircraft only starts flying from the phase-in date, and before that, its usage parameters remain 0.

Operational Constraints

The operational constraints can be divided into two categories: commercial constraints and maintenance constraints. The operations center of airlines defines the commercial constraints. For example, the operations center may limit the availability of the aircraft to perform maintenance during commercial peak seasons (e.g., during the summer or specific holidays), or it may impose an ear-

243 lier time limit to the maintenance check of a specific aircraft following the end
244 of a leasing contract or the chartering of an aircraft to third parties.

245 The maintenance constraints are defined by the maintenance department,
246 which specifies the maintenance capacity according to available maintenance
247 resources, e.g., maintenance tools, workforce, and aircraft spare parts. This
248 capacity is expressed as maintenance slots per day that define how many aircraft
249 can be at the hangar for a specific type of maintenance. Furthermore, other
250 maintenance constraints may apply, such as that no heavy checks can start on
251 the same day to avoid high demanding works in parallel or that some aircraft
252 already have maintenance predefined before computing the schedule. The latter
253 takes place, e.g., when part of the maintenance program is executed by third-
254 parties or partially depends on third-parties, not being subject to rescheduling.
255 A typical example of this is the replacement of landing gears or engines. Besides,
256 maintenance task execution follows the sequence of opening the access panel,
257 inspection, maintenance, and closing the access panel.

258 *Workload of Each Task and Available Workforce*

259 The workload of each task is provided by the airline. Each task associates a
260 task code, a set of skill types required to perform the task, labor hours for each
261 skill type defined by the MPD. If there are urgent unscheduled tasks, they can
262 be added to the input with corresponding duration, workforce, and due dates.

263 The available workforce is the input given by airlines and divided per skill
264 types (e.g., engines and flight control systems, avionics, aircraft metallic struc-
265 ture, and painting, etc.). The available workforce includes the total number of
266 maintenance technicians per skill type, the number of hours a technician work
267 per day on average, and the number of available technicians on each week in
268 the year. The available workforce constrains the task allocation to maintenance
269 checks because it is limited per day, according to the daily workforce schedule.
270 Since aircraft maintenance work is usually ongoing 24 hours every day, airlines
271 divide the daily workforce into three groups of workers to perform their duties
272 and call those groups *morning shift*, *afternoon shift*, and *night shift*. In the

input data, the maintenance planners of airlines have to specify the maximum number of technicians in one shift and also for one task.

3.2. Optimization Model and Algorithms

The model layer has three optimization models in total: a maintenance check scheduling model (AMPO-1 in Figure 1), a maintenance task allocation model (AMPO-2), and a shift planning model (AMPO-3). The design of the model layer follows the top-down approach. The DSS first generates an optimal aircraft maintenance check schedule in AMPO-1, then allocates the maintenance tasks to each maintenance check in AMPO-2. After that, it plans the shifts according to the maintenance tasks to be executed in each letter check.

The reason for following the top-down approach is that it is impossible to plan the work shifts before knowing the task execution or plan all maintenance tasks one after another for the entire fleet without knowing the maintenance check schedule. The maintenance check schedule indicates in which letter check a maintenance task could be allocated without violating the safety regulation defined by the MPD. The work shifts can only be planned based on the maintenance check schedule and the tasks to be executed within each check. The overall optimization process entails the following seven steps:

Step I: Extract maintenance check and task inspection interval

The *Model* component extracts the maintenance check interval and inspection intervals of all tasks from the MPD stored in the database. The inspection intervals are the maximum DY/FH/FC allowed in commercial operation.

Step II: Collect DY/FH/FC of each aircraft and remaining utilization of all systems and components

The *Model* component loads the fleet status (current DY/FH/FC for each check type since its previous execution) and average aircraft daily utilization (FH/day and FC/day) stored in the database (the second step in the model layer of Figure 1). It also collects the usage parameters of all aircraft systems

and components and computes the remaining utilization of each system and component. For example, consider a component of an aircraft with max usage parameters 120 DY, 1000 FH, and 600 FC, and this aircraft has daily utilization of 10 FH/day and 5 FC/day. Given current usage parameters 500 FH and 250 FC, the remaining utilization of this component would be 50 days.

Step III: Identify maintenance opportunities and detailed operational constraints

According to the input constraints from the operation center and maintenance department of airlines, the *Model* component identifies the maintenance opportunities. The maintenance opportunities indicate the time-window when a specific check type is allowed to be performed and the corresponding check capacity. Table 3 presents a format of maintenance opportunities stored in the database after input processing:

Table 3: An example of maintenance opportunities stored in database.

Fleet	Check Type	Start Date	End Date	Capacity
A320	C-/D-Check	Oct-1-2017	May-31-2018	3
A320	A-Check	Every Monday	Every Friday	1
A320	A-Check	Sep-26-2017	Sep-26-2017	2
⋮	⋮	⋮	⋮	⋮

For a specific maintenance check type, if a time window is not within any *Start Date* and *End Date* in Table 3, it means that the associated capacity for this period is 0.

Step IV: Generate optimal aircraft maintenance check schedule (AMPO-1)

After processing and loading the input data, the user can specify the planning horizon for aircraft maintenance check scheduling (AMCS) optimization. The default planning horizon is three years to ensure that it includes at least one C-check for each aircraft, but the user can choose from two to six years.

The model formulation of AMPO-1 can be seen in Appendix A. Currently, there is only one objective function within the DSS for AMCS, minimizing the

323 unused flight hours of the entire fleet [3] for a period specified by the user. It is
 324 possible to add more objectives or even multi-objectives later on. The optimal
 325 letter check schedule is generated using a dynamic programming (DP) based
 326 methodology, as presented in [5]. The idea is to check whether the maintenance
 327 capacity in the future is sufficient or not for each maintenance action (e.g., per-
 328 forming a C-check or several A-checks). This methodology follows a forward
 329 induction approach, incorporating a maintenance priority solution to deal with
 330 the multi-dimensional action vector, as well as a discretization and state aggre-
 331 gation strategy to reduce outcome space at each time stage. If the input data
 332 does not lead to a feasible maintenance check schedule, the DSS will suggest the
 333 best dates for adding necessary maintenance slots to make it feasible.

334 *Step V: Generate optimal task allocation for maintenance checks for each air-*
 335 *craft (AMPO-2)*

336 Once the AMPO-1 plans the optimal letter check schedule for the entire
 337 fleet, the DSS allocates the maintenance tasks to each letter check, assuming
 338 that there are sufficient aircraft spare parts and maintenance tools. The task al-
 339 location aims at minimizing the total cost in task execution, subject to the daily
 340 available workforce. It adopts an algorithm based on the worst-fit decreasing
 341 (WFD) [30]. The task allocation algorithm treats the maintenance resources
 342 within each check as bins and the maintenance tasks as items. It consists of:

- 343 • *Bin Definition:* The task allocation within AMPO-2 divides the entire
 344 aircraft letter check schedule into time segments (bins) according to the
 345 number of parallel maintenance checks. For example, in Figure 2, C1.2,
 346 C12.1, C7.1, C7.2, and C9.1 are the maintenance checks. T1–T7 are the
 347 bins defined by the AMPO-2. The sizes of the bins (time segments) are
 348 determined based on the aircraft maintenance resources, i.e., the number
 349 of maintenance technicians working during the time periods of the bins
 350 (the available workforce per day is given in the input).
- 351 • *Bin Selection:* The heuristic algorithm sorts the time segments according
 352 to the associated capacity (maintenance resources), from highest to lowest.

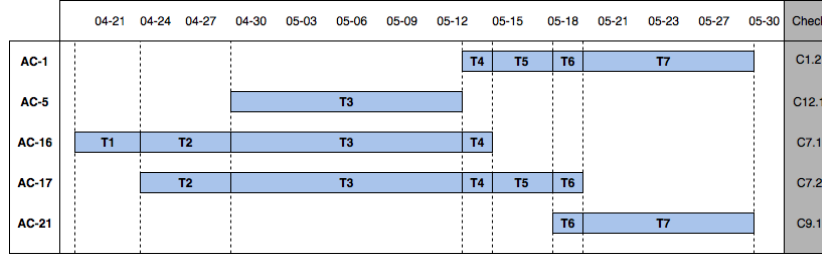


Figure 2: Overlapping maintenance checks are divided into several time segments (bins) in AMPO-2 - i.e., T1, T2, ..., T7.

When the algorithm selects a bin to allocate a maintenance task, it always starts with the bin with the highest remaining capacity. The availability of bin (time-segment) depends on the aircraft having letter checks during that time-segment. In the example of Figure 2, T1 is only available for aircraft (AC) 16, T2 is available for both AC-16 and AC-17, etc.

- *Item Allocation:* The algorithm allocates the items (tasks) following the rules of “the most urgent item (task) first”. Each maintenance task must be allocated before its due date; otherwise, it generates extra capacities and notifies the DSS user.

The model formulation of AMPO-2 is presented in Appendix B, the task allocation algorithm is described in detail in [30]. For the maintenance tasks that have to be executed in a strict order, the task allocation algorithm groups those tasks into a package, and this task package is also considered as one item (one big task). After that, the algorithm allocates the item (task package) to a bin (time segment of a maintenance check). In this way, it ensures that all tasks within the package will be executed. For instance, the maintenance tasks presented in Table 4 have to be executed in the order of:

$$1200-A \longrightarrow 1200-B \longrightarrow 1200-C \longrightarrow 1200-D \quad (1)$$

In this example, technicians have to execute task 1200-A (open the panel at component xxx) first. Otherwise, they cannot continue to inspect or replace the component xxx. After the technicians complete the task 1200-C, they have

Table 4: An example of tasks that have to be executed in the order of $A \rightarrow B \rightarrow C \rightarrow D$.

Fleet	Tail No.	Date	Item	Description
A320	AC-1	Mar-19-2019	1200-A	Open the panel at aircraft component xxx
A320	AC-1	Mar-19-2019	1200-B	Inspect aircraft component xxx
A320	AC-1	Mar-19-2019	1200-C	Replace component xxx
A320	AC-1	Mar-19-2019	1200-D	Close the panel at aircraft component xxx

to execute 1200-D (close the panel at component xxx). The task allocation algorithm groups these four tasks into one package and label it as “Item 1200”, providing information of the sequence when presenting the results to the user.

Step VI: Integrate the optimal maintenance check schedule and task allocation plan

In this step, the DSS first creates a folder for each aircraft with the name “aircraft tail number + Time + Date”, and decouples the entire maintenance check schedule obtained from AMPO-1 according to aircraft tail numbers. In each folder, it saves the associated maintenance checks in the format of Excel. Next, the DSS organizes all the maintenance tasks from AMPO-2 within the same letter check in one table in CSV format and puts this CSV file in the folder according to the aircraft tail number of the letter check. The user can compare or keep track of the historical optimization results according to the time and date in the folder name.

Step VII: Plan the maintenance workforce and shifts (AMPO-3)

The *Model* component also has an algorithm (AMPO-3) to plan the shift (morning/afternoon/night), create job cards, and estimate the workload after the AMPO-2 completes the task allocation for all letter checks. Due to the uncertainty associated with the workforce available per shift in the long term, the optimal maintenance check schedule and task execution plan may quickly

393 become obsolete. Thus, the AMPO-3 only creates the work shifts and job cards
394 for the initial weeks (1–2 weeks) of the planning horizon.

395 The shift planning algorithm allocates the maintenance tasks to each shift,
396 respecting the workforce available per shift and the sequence of opening ac-
397 cess panel, inspection, maintenance, and closing access panel (and this is the
398 only task execution sequence we have to follow in both AMPO-2 and AMPO-3
399 according to the specification of our airline partner). Figure 3 illustrates the
400 workflow of shift planning function (AMPO-3). AMPO-3 first assigns the tasks
401 of opening the access panel to the morning shift. If there is no available work-
402 force left in the morning shift, it continues to assign those tasks to the afternoon
403 shift (or even night shift) until all the tasks of opening access panels are allo-
404 cated. Next, the algorithm assigns the inspection works, and after that, the
405 maintenance tasks. The tasks of closing the access panel are allocated at last.
406 The shift planning process continues until it loops over the task execution plans
407 of all maintenance checks. When it finishes, the DSS will store the results in
408 the database according to the aircraft tail number of the tasks.

409 3.3. Graphical User Interface

410 The GUI serves the purpose of interacting with DSS users. The DSS users
411 can load input data, start the AMPO, visualize the optimization results and
412 associated KPIs, change operational constraints (planning horizon, the number
413 of maintenance slots, or reserve slots for specific maintenance activities), and
414 export the output data via the GUI. Those actions are the basic requirements
415 for the GUI from the DSS users, identified by the AIRMES project group.

416 The GUI of the DSS has a single main window, divided into five screens.
417 The user can see the maintenance check schedule of all aircraft for the entire
418 planning horizon on a daily basis or per hangar view on different screens. The
419 GUI also displays key performance indicators (KPIs), the tasks allocated per
420 maintenance check, the workforce assigned per day (of the first few weeks), the
421 identification of the maintenance interval tolerances used, the maintenance slots
422 generated as additional to the given capacity. The user can also use the GUI to

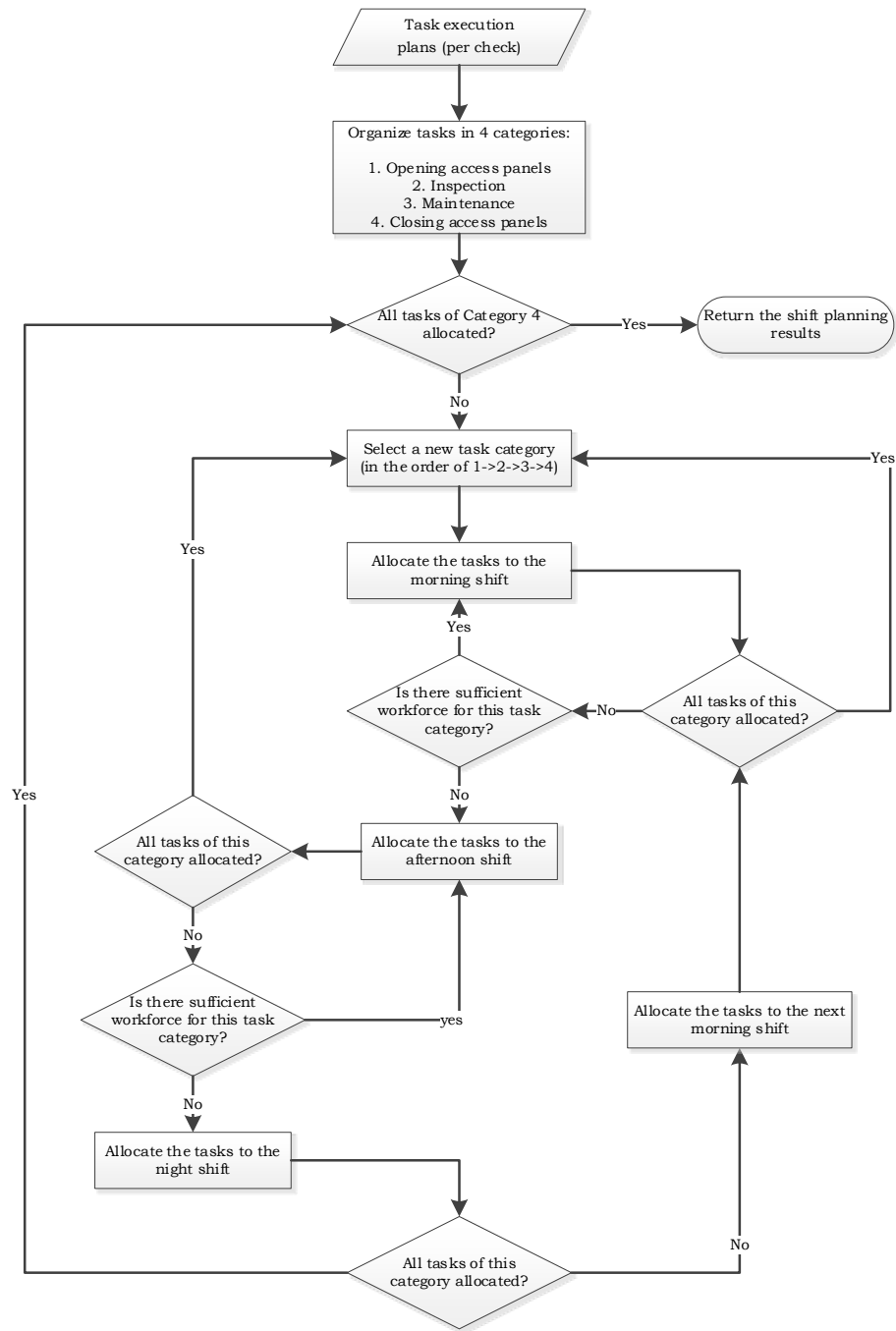


Figure 3: Workflow of shift planning (AMPO-3).

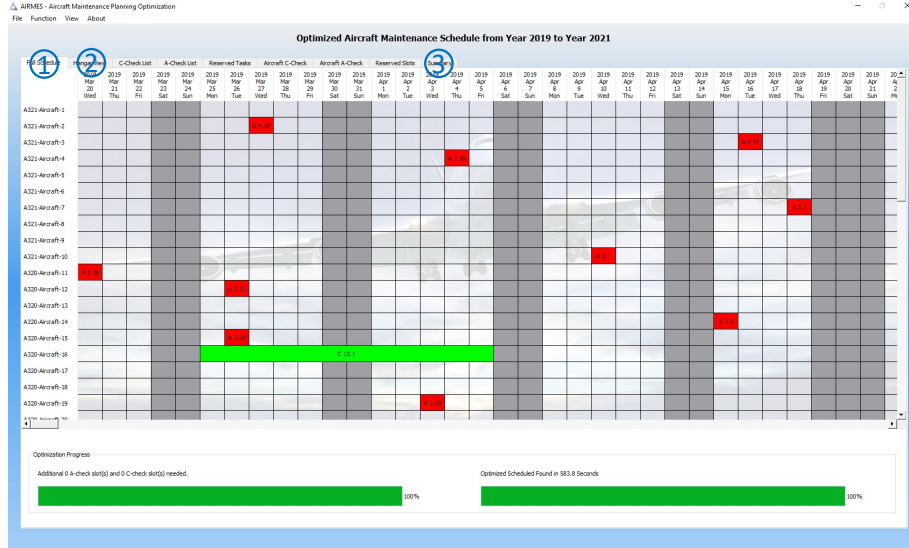


Figure 4: Main screen of the DSS.

set the planning horizon, modify the start dates of specific maintenance checks, and change the operational constraints, such as adding/reducing maintenance slots or reserve maintenance slots for other maintenance activities. We use Figure 4 and 5 to illustrate the main features of the DSS.

In Figure 4, the marker ① indicates the main screen of the DSS. The main screen displays the aircraft maintenance check schedule per day per aircraft, computation time, and the number of extra maintenance slots created during optimization for a specific planning horizon. The marker ② indicates the 2nd screen of the DSS. The 2nd screen displays the maintenance check schedule for the entire fleet in the hangar view. The marker ③ indicates the screen of displaying the KPIs, including the mean FH, mean FC, total maintenance checks, distribution of unused FH and FC for each check type, and the number of merged A- in C-/D-Checks.

In the 2nd screen, the DSS user can further see the maintenance tasks of each check. If the DSS user selects a maintenance check, a dialogue box will be

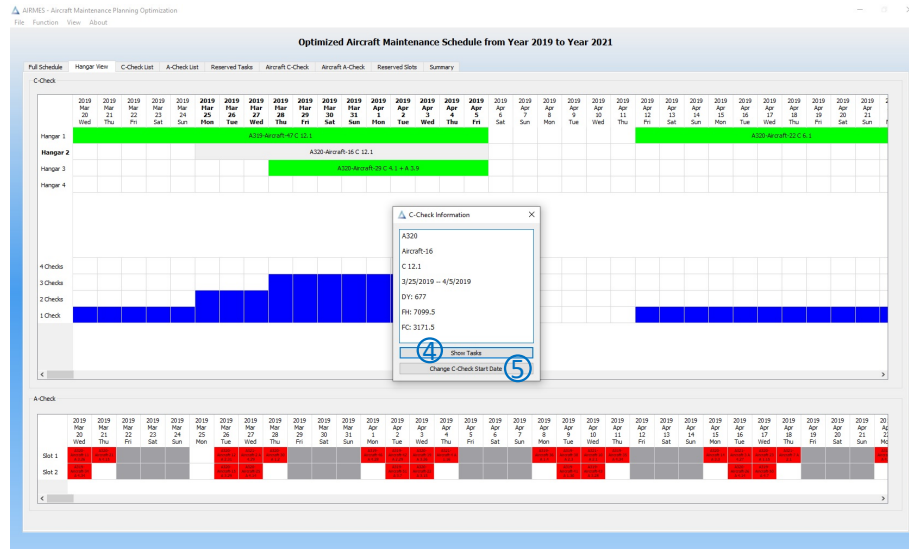


Figure 5: The 2nd screen of the DSS.

438 popped up to display the aircraft tail number, maintenance check label, current
 439 DY, FH, and FC. The user can click the button “Show Tasks”, as indicated
 440 by marker ④ in Figure 5. The DSS will display a list of maintenance tasks
 441 within the check and a figure that shows the workload distribution and the work
 442 shifts. The user can also change the start date of a specific check by clicking the
 443 button indicated by marker ⑤. The DSS will re-optimize the entire schedule
 444 according to the new specification from the user.

445 4. Demonstration and Evaluation

446 The DSS was demonstrated and validated in a demonstration exercise or-
 447 ganized as part of the AIRMES project, on 51 aircraft, in March 2019. The
 448 exercise was carried out in collaboration with one of the major European air-
 449 lines and one of the leading aircraft manufacturers and observed by the Clean
 450 Sky 2 Joint Undertaking partners involved in the related research effort.

451 This exercise aimed to validate the value of the DSS and demonstrate its ap-

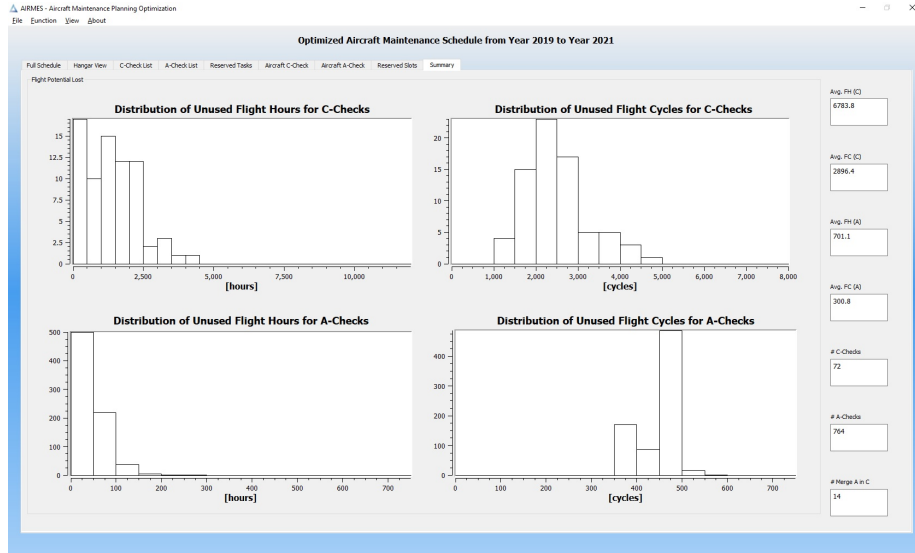


Figure 6: The KPIs of maintenance check schedule of the airline. We used the DSS to load the maintenance check schedule of the airline directly and visualized the results on the interface.

452 plicability, primarily for AMP optimization and the study of future maintenance
 453 scenarios. For this reason, two test cases were performed and discussed. In the
 454 first test case, we aimed to validate the DSS and benchmark its performance by
 455 comparing the solution obtained with the maintenance schedule of the airline.
 456 In the second test case, we investigated the current considerations of the airline
 457 about its future maintenance policies and fleet developments. The results were
 458 checked and validated by the airline experts, providing valuable insights to the
 459 airline on future maintenance limitations and solutions.

460 4.1. Standard Aircraft Maintenance Check Scheduling Optimization on Fleet 461 Maintenance Data

462 We received the input for AMCS on March 19th 2019 and optimized the A-
 463 and C-checks for the A320 family of our airline partner from March 20th 2019
 464 to December 31st 2021, under the same operational constraints as the airline.
 465 According to the requirements of our airline partner, D-checks are merged within

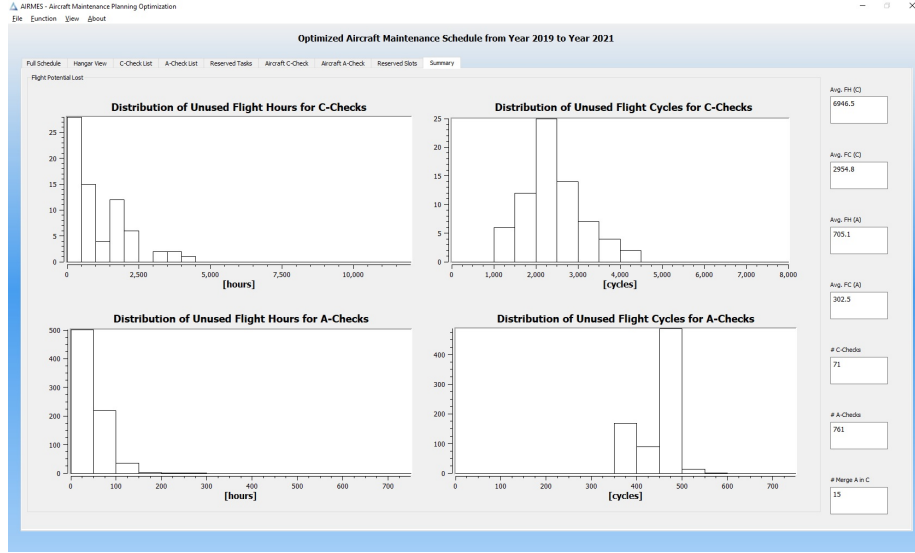


Figure 7: The KPIs of maintenance check schedule from the AMPO-1 of the DSS.

C-check in the following pattern:

$$C-1, \quad C-2, \quad \underbrace{C-3}_{D\text{-check}}, \quad C-4, \quad C-5, \quad \underbrace{C-6}_{D\text{-check}}, \quad C-7, \quad C-8, \quad \underbrace{C-9}_{D\text{-check}}, \dots \quad (2)$$

We compared our results with the maintenance schedule available at the airline (Airline Schedule). According to the results illustrated in Figure 6 and 7, the AMPO-1 of the DSS outperforms the planning approach of the airline. The AMPO-1 results in 6946.5 FH for C-check and 705.1 FH for A-check, higher than 6783.8 FH and 701.1 FH from the maintenance schedule of the airline, but the result of AMPO-1 has one fewer C-check and three fewer A-checks. Our airline partner also checks the maintenance check schedule obtained using the DSS and agrees that the DSS generates a better schedule than the maintenance planners. Besides, the AMPO-1 of the DSS optimizes both the aircraft A- and C-check schedule for 2019–2021 within only 10 minutes. It means that the DSS user can run the DSS to update its aircraft maintenance check schedule if there are changes instead of manually shuffling the A-/C-checks to make another feasible one.

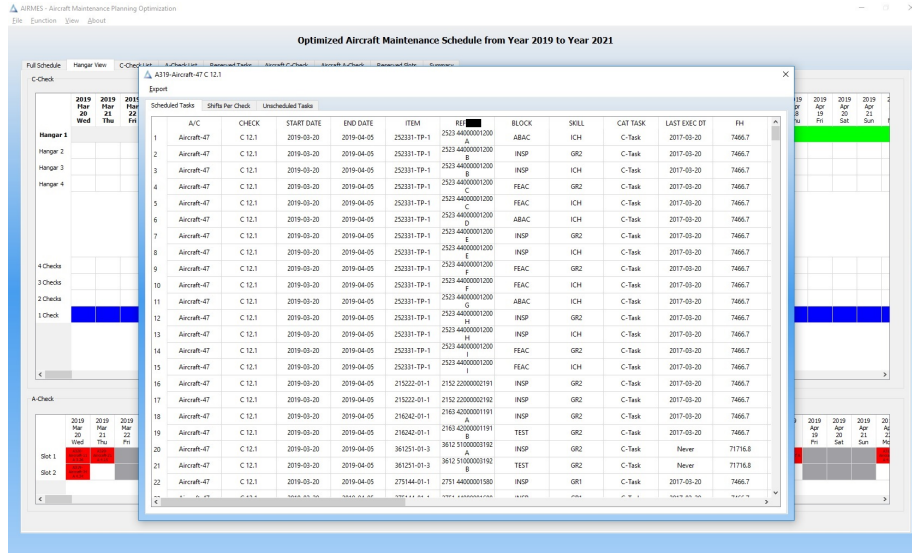


Figure 8: Results of optimal task allocation (AMPO-2).

From a saving and revenue management perspective, since airlines spend on average \$150K–\$350K on a C-check [1] and \$10K–\$15K on an A-check, one fewer C-check and three fewer A-checks in total can result in a potential saving of \$0.1M–\$0.4M for the considered time horizon of roughly three years. Furthermore, a C-check lasts about 1–4 weeks, and an A-check lasts 24 hours in this case study. One reduced C-checks and three fewer A-checks are equivalent to about 10–31 days of aircraft availability for commercial operations. This may generate a considerable amount of revenue for the airline. According to an economic evaluation performed by another Clean Sky project, the implementation of the DSS can potentially reduce the base maintenance costs by 2.7% for point-to-point carrier airlines, 0.5% for large hub and spoke carrier airlines, and 2.4% for small hub and spoke carrier airlines.

Following the demonstration of AMPO-1, we continued to test the optimal task allocation (AMPO-2) of the DSS. The AMPO-2 allocated the maintenance tasks using the optimal maintenance check schedule created by the AMPO-1. It

Aircraft	Date	Shift	Item	Block	Skill Type and Required Labor-Hours							
					S 1	S 2	S 3	S 4	S 5	S 6	S 7	S 8
AC-1	09-20-2018	Morning	Open panel '734'	—	0.6	0.2	0	0	0	0	0	0
AC-1	09-20-2018	Morning	Open panel '151KW'	—	0	0.2	0	0	0	0	0	0
AC-1	09-20-2018	Morning	Open panel '312AR'	—	0.1	0	0	0	0	0	0	0
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
AC-1	09-20-2018	Afternoon	251100-TP-1	INSP	0	0	0	0	2.0	0	0	0
AC-1	09-20-2018	Afternoon	575161-01-2	INSP	0.8	0	0	0.2	0.2	0.2	0.1	0
AC-1	09-20-2018	Afternoon	792000-C4-2	LUB	0.5	0	0	0	0	0	0	0
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
AC-1	09-20-2018	Night	575164-01-4	ABAC	0.5	0	0	0	0	0	0	0
AC-1	09-20-2018	Night	575164-01-4	INSP	0	0	0	0	0	0	0	0.5
AC-1	09-20-2018	Night	575164-01-4	FEAC	0.5	0	0	0	0	0	0	0
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

Figure 9: An example of work shifts planned by AMPO-3.

addressed the task allocation for maintenance check schedule of Mar 20th 2019–
Dec 31st 2021 within 10 minutes. The outcome of AMPO-2 is an optimal task
allocation plan for the entire fleet and all letter checks, including over 60,000
tasks. An example of the outcome from AMPO-2 can be seen in Figure 8. To
verify the AMPO-2, we compared the optimal task allocation plan with the
results from a commercial optimization solver. The comparison showed that
the AMPO-2 produces results within an optimality gap of only 0.028%. Our
airline partner also validated the optimal task allocation plan and its feasibility
by benchmarking our solution with the task allocation solution they had for the
following year. The maintenance planners of the airline stated that the results
from optimal task allocation are feasible for practical implementation.

After the AMPO-2 completed the optimal task allocation for all mainte-
nance checks, the AMPO-3 planned the work shifts and creates job cards for
technicians. Our airline partner set the horizon of shift planning for two weeks.
Figure 9 shows an example of the results from AMPO-3. The 1st column shows
the aircraft tail number. The 2nd and 3rd columns indicate the date and work

shifts. The 4th column describes the item or action, and the 5th column tells the maintenance planner where the maintenance work is in the aircraft. The last eight columns imply the workload needed for each skill type. The airline evaluated the work shifts after the demonstration and indicated that the work shifts of the first 2–3 days are almost the same as they planned, yet the difference becomes dramatic in the second week. It is worth mentioning that, for example, if a task requires one labor-hour for a specific skill type, it can be one technician spending an hour, or two technicians spend half an hour, or even four technicians spend 15 minutes performing the task.

4.2. Evaluation of Aircraft Maintenance Strategies using the DSS

In the second test case, we used the DSS to evaluate different maintenance strategies before implementation. Each strategy is modeled as a test scenario, and all scenarios are compared to the baseline scenario (Airline Schedule). Three maintenance strategies (test scenarios) were proposed by the airline:

1. *Scenario 1*: increase the number of daily C-check slots from three to four but shorten the period in which C-checks can be performed from the current *October–May* to *November–March*;
2. *Scenario 2*: increase the fleet size from 51 to 66 aircraft without changing the maintenance periods or number of slots available;
3. *Scenario 3*: increase the fleet size from 51 to 66 aircraft but now increasing the A-check slots by one on Fridays.

Table 5 shows the KPIs from test scenarios. We also include the KPIs from the previous demonstration and use the airline schedule as the baseline scenarios. First of all, without considering other costs, we see the benefit per aircraft from implementing the DSS (DSS Schedule) compared with the *baseline* scenario (Airline Schedule) for a 3-year planning horizon:

$$\text{C-Check:} \quad \underbrace{20.6}_{\text{gain}} + \underbrace{4.9}_{\text{saving}} - \underbrace{0}_{\text{cost}} = 30.9\text{K} \quad (3)$$

$$\text{A-Check:} \quad \underbrace{5.7}_{\text{gain}} + \underbrace{0.7}_{\text{saving}} - \underbrace{0}_{\text{cost}} = 6.4\text{M} \quad (4)$$

Table 5: Summary of KPIs from the airline schedule (3rd column), the AMCS optimization for the first test case (4th column), and the different scenarios from the second test case (5th–7th column). The “Airline Schedule” serves as the baseline scenario. “Gain” represents the potential income generated per aircraft from having more days for commercial operations (due to more/few days for A- or C-checks) compared with the baseline scenario. “Saving” represents the reduction of maintenance costs per aircraft due to more/fewer checks. “Cost” represents the costs per aircraft for creating extra slots.

KPIs 20/03/2019–31/12/2021		Airline Schedule	DSS Schedule	Scenario 1	Scenario 2	Scenario 3
C-check	Average FH	6783.8	6946.5	6959.8	6543.8	6012.3
	Average FC	2896.4	2954.8	2955.4	2802.5	2570.2
	Total Checks	72	71	69	73	76
	Extra Slots	0	0	61	0	0
	Gain [\$]	—	26.0K	78.0K	-20.1K	-80.4K
	Saving [\$]	—	4.9K	14.7K	-3.8K	-15.2K
	Cost [\$]	—	0	66.0K	0	0
A-check	Average FH	701.1	705.1	665.0	690.6	664.3
	Average FC	300.8	302.5	285.6	292.7	281.6
	Total Checks	764	761	811	929	967
	Extra Slots	3	3	4	75	9
	Gain [\$]	—	5.7K	-89.9K	-243.8K	-299.9K
	Saving [\$]	—	0.7K	-11.5K	-31.3K	-38.4K
	Cost [\$]	—	0	0.7K	40.9K	3.4K
Total Benefit per Aircraft		—	37.3K	-75.4K	-339.8K	-437.3K

According to our airline partner:

- 1) One day of operation generates on average \$97.5K of revenue.
- 2) The A-check of an A320 family aircraft lasts one working day and costs on average \$12.5K.
- 3) The C-check of an A320 family aircraft lasts on average 13.6 working days (slots).
- 4) One fewer A-(C-)check means the entire fleet can have 1(13.6) more days for operations.
- 5) The C-check of an A320 family aircraft costs on average \$250K (\$18.4K per working day).
- 6) The cost of creating one extra A-/C-check slot is three times as one normal slot.

537 The KPIs of *Scenario 1* indicate that shortening the C-check periods while
 538 increasing the C-check capacity, as considered by the airline, is not enough
 539 to cope with the C-check demand from the current fleet size and leads to a
 540 loss of \$75.4K in total per aircraft for two check types. Although it gains
 541 \$78.0K from more days of commercial operations compared with the *baseline*
 542 scenario (Airline Schedule) and saves \$14.7K because of performing fewer C-
 543 checks, the airline needs to spend \$66.0K per aircraft on creating extra C-check
 544 slots (non-existent daily slots are added to the schedule, representing moments
 545 that technicians have to work extra-time or that additional workforce has to
 546 be hired). The majority of loss comes from A-check due to grounding aircraft
 547 more often for A-checks (-\$89.9K from commercial operations and -\$11.5K for
 548 performing more A-check) and creating more extra A-check slots (-\$0.7K). The
 549 reason is that the optimization algorithm of AMPO-1 tries to ground the aircraft
 550 for A-check more often to defer the need for a C-check. For example, consider
 551 an aircraft with a C-check interval of 7500 FH/730 DY and an average daily
 552 operation of 15 FH. If this aircraft has no A-check, it will be grounded and
 553 performed a C-check after 500 days since the FH usage parameter reaches 7500
 554 FH before the DY usage parameter reaches 730 DY. If there is one A-check
 555 scheduled before the C-check (A-check lasts one day), this aircraft can have
 556 the C-check after 501 days. Similarly, if the aircraft is scheduled two A-checks,
 557 it can have the C-check after 502 days, and so forth. Based on the results of
 558 the *Scenario 1* evaluation, we suggested that the airline should use its current
 559 maintenance strategy rather than the new one (described in *Scenario 1*).

560 For *Scenario 2*, we observed that the current A-check capacity is not suffi-
 561 cient to handle the increased A320 fleet size according to the results shown in
 562 Table 5. Eventually, it leads to a huge loss, -\$339.8K, on average per aircraft.
 563 Apart from the loss from C-check due to fewer commercial operations (-\$20.1K)
 564 and performing more C-check (-\$3.8K), A-check contributes most to the loss,
 565 -\$243.8K from commercial operations and -\$31.3K from performing more A-
 566 checks. Besides, it needs to spend \$40.9K on average per aircraft on creating
 567 extra A-check slots.

568 To cope with the soaring A-check demand from increased fleet size, the air-
 569 line proposes to add one A-check slot on Friday, as described in *Scenario 3*.
 570 According to the DSS results, creating one additional aircraft A-check slot on
 571 Friday (*Scenario 3*) significantly reduces the need for extra A-check capacity
 572 from 75 to 9 compared with *Scenario 2*, meaning that the cost of creating extra
 573 slots is reduced (from \$40.9K to \$3.4K). However, it also increases the number
 574 of checks for both check types, resulting in a huge revenue loss from commercial
 575 operations. The total loss increases by $437.3 - 339.8 = 97.5$ K on average per
 576 aircraft compared with *Scenario 2*. We found out that the optimization algo-
 577 rithm schedules C-check more frequently to provide more opportunities to merge
 578 the A-checks in C-checks (since the airline primarily wanted to avoid creating
 579 extra slots). Based on the results of *Scenario 2* and *Scenario 3*, we suggested
 580 that adding one A-check slot per week is not sufficient for the increase of fleet
 581 size, and the airline should consider adding more A-check slots to cope with the
 582 increased maintenance check demand.

583 **5. Conclusion**

584 This paper presents a novel decision support system (DSS) that addresses
 585 aircraft maintenance planning optimization in an integrated fashion, automat-
 586 ing repetitive tasks while enabling fast, efficient, human-in-the-loop decision
 587 making for optimized planning purposes. First of all, the DSS is capable of
 588 optimizing the aircraft maintenance check schedule. Secondly, based on the
 589 optimal maintenance check schedule, the DSS allocates maintenance tasks to
 590 each maintenance check considering the overlapping situation (having multiple
 591 checks on going in the same period). Thirdly, the DSS plans the work shift re-
 592 specting the task sequence in practice. It can potentially help airlines improve
 593 their maintenance planning efficiency, reduce the related maintenance opera-
 594 tion costs, and even assess their maintenance strategies. Therefore, the DSS
 595 makes significant contributions relevant to both scientific research and industry
 596 application.

597 The DSS bridges the gap between long-term AMCS and short-term shift
 598 planning. It integrates aircraft maintenance check scheduling, maintenance task
 599 allocation and, work shift planning in the same platform. A demonstration
 600 exercise with a European airline shows that the DSS can generate a compre-
 601 hensive optimal maintenance plan for a planning horizon of three years within
 602 half an hour. It means airlines can use the DSS to reduce the time needed for
 603 aircraft maintenance planning from several days to about 30 minutes. More
 604 importantly, considering the uncertainty that might impact aircraft utilization
 605 or maintenance activities, we make it possible for the maintenance planners to
 606 run the DSS in a short time to update the current plan. Whenever there are
 607 changes in the aircraft maintenance tasks or maintenance activities, mainte-
 608 nance planners can quickly make new decisions using the DSS and re-organize
 609 the tools, workforce or promptly prepare the aircraft spare parts.

610 Besides, the demonstration exercise results show that the DSS reduces the
 611 number of A-/C-checks by three/one while increasing the expected average FH
 612 of A-/C-check by 2.4%/0.6% for a planning horizon of 3 years compared with
 613 the maintenance check schedule made using the planning approach of the airline.
 614 The reduced A- and C-checks could lead to a significant saving in maintenance
 615 costs. The improved aircraft utilization also indicates that there will be more
 616 aircraft available for day-to-day commercial operations to generate additional
 617 revenue for airlines. After the demonstration exercise, the DSS was tested and
 618 classified by the Clean Sky Joint Undertaking partners to be at a Technology
 619 Readiness Level Six (TRL 6). Nevertheless, the tool still has some limitations
 620 that have to be addressed in the future if a higher TRL is aimed:

- 621 • The primary goal of AIRMES was on the development of the optimization
 622 algorithms, so future efforts should focus on improving the GUI.
- 623 • Define requirements and specifications that will facilitate direct integration
 624 of the DSS with other information systems used by airlines, including
 625 the development of the Application Programming Interface (API) and,
 626 potentially, a Software Development Kit (SDK).

- Include the number of aircraft spare parts in the constraints in the task allocation (AMPO-2) model.

Another interesting direction is to incorporate condition-based maintenance (CBM) by taking health prognostics and diagnostics into consideration when developing maintenance plans. Although including CBM in the DSS will increase model complexity and computation time, it will prepare the tool to cope with a current trend in the aircraft maintenance research and operational communities.

Finally, it is worth mentioning that although the DSS is tailored to aircraft maintenance planning optimization, it can also be adjusted to address similar problems, such as train or bus maintenance planning for the coming days or weeks, or to match the maintenance demand with operation timetables. For example, the main screen of the DSS can be changed to display daily operation hours and maintenance duration. The algorithm described in [5] can be adapted for similar maintenance scheduling or even more general scheduling problems (e.g., vehicle routing or production planning) since the idea of the algorithm is to estimate the consequence of each possible (maintenance) action before making a decision. For such applications, the DSS framework can be maintained.

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765 343.

766 **Appendix A. AMPO-1 Model Formulation**

767 *Aircraft Maintenance Check Scheduling Model*

768 In Appendix A, we present the model formulation of AMPO-1 (aircraft
769 maintenance check scheduling) of the proposed DSS and briefly explain the
770 objective function and constraints. A detailed description of the aircraft main-
771 tenance check scheduling (AMCS) model and the associated optimization algo-
772 rithm can be found in [5].

$$\min_{\pi} \sum_{t=t_0}^T \sum_{k \in K} \sum_{i=1}^N \left[\chi_{i,t}^k \left(I_{k\text{-FH}}^i - \text{FH}_{i,t}^k \right) + (1 - \chi_{i,t}^k) P_a \theta_{i,t}^k + P_d \eta_{i,t}^k \right] \quad (\text{A.1})$$

773 subject to:

$$z_{i,t}^k = \begin{cases} 0 & \text{if } z_{i,t}^k = t - 1 \\ z_{i,t}^k & \text{otherwise} \end{cases} \quad (\text{A.2})$$

$$\tilde{\delta}_{i,t}^k = \begin{cases} 0 & \text{if } z_{i,t}^k = t - 1 \\ 1 & \text{otherwise} \end{cases} \quad (\text{A.3})$$

$$\tilde{M}_t^k = \sum_h M_{h,t}^k - \sum_{i=1}^N \tilde{\delta}_{i,t}^k \quad (\text{A.4})$$

$$z_{i,t+1}^k = \begin{cases} t + L_i^k(y_{i,t}^k) & \text{if } \chi_{i,t}^k = 1 \\ \tilde{z}_{i,t}^k & \text{otherwise} \end{cases} \quad (\text{A.5})$$

$$\delta_{i,t+1}^k = \begin{cases} 0 & \text{if } \chi_{i,t}^k = 1 \\ \tilde{\delta}_{i,t}^k & \text{otherwise} \end{cases} \quad (\text{A.6})$$

$$\Delta \text{DY}_{i,t+1}^k = \underbrace{(\text{DY}_{i,t}^k + 1)}_{\text{DY}_{i,t+1}^k} - \underbrace{[I_{k\text{-DY}}^i + (1 - \theta_{i,t}^k) e_{k\text{-DY}}^i - \epsilon_{i,t}^{k\text{-DY}}]}_{\text{actual DY interval of type } k \text{ check}} \quad (\text{A.7})$$

779

$$\Delta\Psi_{i,t+1}^k = \underbrace{(\Psi_{i,t}^k + \psi_{i,t})}_{\text{usage parameters of } t+1} - \underbrace{[I_{k-\Psi}^i + (1 - \theta_{i,t}^k) e_{k-\Psi}^i - \epsilon_{i,t}^{k-\Psi}]}_{\text{actual interval of } \Psi \ (\Psi \in \{\text{FH}, \text{FC}\})} \quad (\text{A.8})$$

780

$$\eta_{i,t+1}^k = \begin{cases} 1 & \chi_{i,t}^k = 0, \max\{\Delta\text{DY}_{i,t+1}^k, \Delta\text{FH}_{i,t+1}^k, \Delta\text{FC}_{i,t+1}^k\} > 0 \\ \tilde{\eta}_{i,t}^k & \text{otherwise} \end{cases} \quad (\text{A.9})$$

781

$$\text{DY}_{i,t+1}^k = (1 - \delta_{i,t}^k) (\text{DY}_{i,t}^k + 1) \quad (\text{A.10})$$

782

$$\Psi_{i,t+1}^k = (1 - \delta_{i,t}^k) \left[\Psi_{i,t}^k + (1 - \delta_{i,t}^{K \setminus k}) \psi_{i,t} \right] \quad (\text{A.11})$$

783

$$\epsilon_{i,t+1}^{k-\text{DY}} = \begin{cases} \max\{0, \Psi_{i,t}^k - I_{k-\text{DY}}^i\} & \text{if } \chi_{i,t}^k = 1 \\ \epsilon_{i,t}^{k-\text{DY}} & \text{otherwise} \end{cases} \quad (\text{A.12})$$

784

$$\epsilon_{i,t+1}^{k-\Psi} = \begin{cases} \max\{0, \Psi_{i,t}^k - I_{k-\Psi}^i\} & \text{if } \chi_{i,t}^k = 1 \\ \epsilon_{i,t}^{k-\Psi} & \text{otherwise} \end{cases} \quad (\text{A.13})$$

785

$$\theta_{i,t+1}^k = \begin{cases} 1 & \text{if } \max\{\epsilon_{i,t+1}^{k-\text{DY}}, \epsilon_{i,t+1}^{k-\text{FH}}, \epsilon_{i,t+1}^{k-\text{FC}}\} > 0 \\ 0 & \text{otherwise.} \end{cases} \quad (\text{A.14})$$

786

$$y_{i,t+1}^k = \begin{cases} y_{i,t}^k + 1 & \text{if } \chi_{i,t}^k = 1 \\ y_{i,t}^k & \text{otherwise} \end{cases} \quad (\text{A.15})$$

787

$$L_i^k(y_{i,t+1}^k) \text{ is the elapsed time specified by airline.} \quad (\text{A.16})$$

$$\text{fh}_{i,t+1}^k \text{ is estimated according to historical aircraft FH.} \quad (\text{A.17})$$

$$\text{fc}_{i,t+1}^k \text{ is estimated according to historical aircraft FC.} \quad (\text{A.18})$$

788

$$\text{DY}_{i,t+1}^k \leq I_{k-\text{DY}}^i + (1 - \theta_{i,t}^k) e_{k-\text{DY}}^i - \epsilon_{i,t}^{k-\text{DY}} \quad (\text{A.19})$$

789

$$\Psi_{i,t+1}^k \leq I_{k-\Psi}^i + (1 - \theta_{i,t}^k) e_{k-\Psi}^i - \epsilon_{i,t}^{k-\Psi} \quad (\text{A.20})$$

$$\chi_{i,t}^k \leq \frac{\sum_{\tau=t}^{t+L_i^k(y_{i,t}^k)} M_{h,\tau}^k}{L_i^k(y_{i,t}^k)}, \quad t \in [t_0, T] \quad (\text{A.21})$$

$$\sum_{i=1}^N \delta_{i,t}^k \leq \sum_h M_{h,t}^k, \quad t \in [t_0, T] \quad (\text{A.22})$$

$$\sum_{i=1}^N \chi_{i,t}^k \leq \begin{cases} 1 & \text{if } d_k > 0 \text{ and } \sum_{i=1}^N \chi_{i,\tau}^k = 0, \forall \tau \in [t - d_k, t) \\ M_t^k & \text{otherwise} \end{cases} \quad (\text{A.23})$$

The objective of the aircraft maintenance check scheduling is to minimize the sum of the total unused flight hours $\chi_{i,t}^k (I_{k\text{-FH}}^i - \text{FH}_{i,t}^k)$ of all maintenance checks and for the entire fleet over the planning horizon $[t_0, T]$. P_a is the penalty of using maintenance interval tolerance (using tolerance is allowed when there is no feasible maintenance check schedule found during the optimization). P_d is the cost of creating an extra maintenance slot.

Constraints (A.2)–(A.6) update the available maintenance slots of all letter check types before scheduling aircraft letter checks. Constraints (A.7)–(A.18) update the utilization of each aircraft, depending on the maintenance check decision of day t (how many aircraft will be scheduled maintenance checks and how many checks should be performed for each check type). Constraints (A.19)–(A.23) ensure that the number of maintenance check scheduled are no larger than the available slots.

We proposed a dynamic programming (DP) based methodology to address the AMCS. First of all, we find out how many possible actions (i.e., how many checks can be performed for each check type) of a day. Secondly, for each possible action, we calculate if the number of available slots in the future is sufficient after performing the action. Among the possible actions, we only keep the ones that require minimal extra slots and the associated fleet status after performing the action and then move forward to the next day using the new fleet status. The detail of the DP-based methodology can also be found in [5].

AMCS Model Parameters:

d_k	Minimum interval between the start dates of two type k checks.
e_{k-DY}^i	Maximum DY tolerance of type k check interval of aircraft i
e_{k-FH}^i	Maximum FH tolerance of type k check interval of aircraft i
e_{k-FC}^i	Maximum FC tolerance of type k check interval of aircraft i
$fc_{i,t}$	Average daily FC usage for aircraft i at day t
$fh_{i,t}$	Average daily FH usage for aircraft i at day t
I_{k-DY}^i	Interval of type k check of aircraft i in terms of DY
I_{k-FH}^i	Interval of type k check of aircraft i in terms of FH
I_{k-FC}^i	Interval of type k check of aircraft i in terms of FC
P_d	Cost for creating one extra maintenance slot
P_a	Penalty for an aircraft using the tolerance

Other Parameters:

h	Hangar indicator
i	Aircraft indicator
k	Maintenance check type indicator, $k \in K = \{A, B, C, D\}$
N	Total number of aircraft
n_k	The number of hangars for type k check
t	Indicator of calendar day
T	Final day in planning horizon
t_0	First day in planning horizon

Decision Variables and Related Attributes:

$DY_{i,t}^k$	Total DY of aircraft i in the beginning of day t for type k check
$FC_{i,t}^k$	Cumulative FC of aircraft i at t since last type k check
$FH_{i,t}^k$	Cumulative FH of aircraft i at t for type k check
$L_i^k(y_{i,t}^k)$	Estimated elapsed time of next type k check with label $y_{i,t}^k$
$M_{h,t}^k$	Binary variable to indicate if a type k check can be performed in hangar h on the day t
M_t^k	Hangar capacity of type k check, $M_t^k = \sum_h M_{h,t}^k$
$y_{i,t}^k$	Next maintenance label for of type k check of aircraft i on the day t
$z_{i,t}^k$	The end date of type k check of aircraft i
$\delta_{i,t}^k$	Binary variable to indicate if aircraft i is undergoing a type k check on the day t
$\epsilon_{i,t}^{k-DY}$	Extra DY before day t if previous type k check is deferred
$\epsilon_{i,t}^{k-FH}$	Extra FH before day t if previous type k check is deferred
$\epsilon_{i,t}^{k-FC}$	Extra FC before day t if previous type k check is deferred
$\eta_{i,t}^k$	Binary variable to indicate if aircraft i is grounded and waiting for a slot of type k check on the day t
$\theta_{i,t}^k$	Tolerance usage indicator of type k check of aircraft i on the day t
$\chi_{i,t}^k$	Binary variable to indicate if aircraft i starts a type k check on the day t
x_t	Available action on the day t
π	Scheduling policy
Ψ	$\Psi \in \{FH, FC\}$
$\Psi_{i,t}^k$	$\Psi_{i,t}^k \in \{FH_{i,t}^k, FC_{i,t}^k\}$
$\psi_{i,t}^k$	$\psi_{i,t}^k \in \{fh_{i,t}^k, fc_{i,t}^k\}$

816 Appendix B. AMPO-2 Model Formulation

817 Aircraft Maintenance Task Allocation Model

818 In Appendix B, we present the model formulation of AMPO-2 (aircraft
819 maintenance task allocation for each letter check) of the DSS and briefly ex-
820 plain the objective function and constraints. A detailed description of the task
821 allocation model and the associated solution approach can be found in [30].

$$\min \sum_{k \in K} \sum_{i \in N_k} \sum_{t \in R_{i,k}} c_{i,k}^t \cdot x_{i,k}^t \quad (\text{B.1})$$

822 subject to:

$$\sum_{t \in R_{i,k}} x_{i,k}^t = 1, \quad \forall i \in N_k, \quad \forall k \in K \quad (\text{B.2})$$

$$\sum_{k \in K} \sum_{i \in N_k} \sum_{j \in J} GR_{i,k}^j \cdot x_{i,k}^t \cdot \sigma^{j,l} \leq \overline{GR}_t^l, \quad \forall t \in T_k, \quad \forall l \in J \quad (\text{B.3})$$

$$\sum_{m \in R_{p,k}} d^m \cdot x_{p,k}^m - \sum_{t \in R_{i,k}} d^t \cdot x_{i,k}^t \leq \bar{d}_{i,k}, \quad \forall i \in N_k, \quad \forall p \in O_{i,k}, \quad \forall k \in K \quad (\text{B.4})$$

$$\sum_{m \in R_{p,k}} fh^m \cdot x_{p,k}^m - \sum_{t \in R_{i,k}} fh^t \cdot x_{i,k}^t \leq \bar{fh}_{i,k}, \quad \forall i \in N_k, \quad \forall p \in O_{i,k}, \quad \forall k \in K \quad (\text{B.5})$$

$$\sum_{m \in R_{p,k}} fc^m \cdot x_{p,k}^m - \sum_{t \in R_{i,k}} fc^t \cdot x_{i,k}^t \leq \bar{fc}_{i,k}, \quad \forall i \in N_k, \quad \forall p \in O_{i,k}, \quad \forall k \in K \quad (\text{B.6})$$

$$x_{i,k}^t \in \{0, 1\}, \quad \forall k \in K, \quad \forall i \in N_k, \quad \forall t \in T_k \quad (\text{B.7})$$

828 The objective function (B.1) aims at minimizing the total de-escalation costs,
829 which can be calculated by comparing how earlier the task item is allocated
830 when compared with its desired due day [31]:

$$c_{i,k}^t = \frac{O_day_i - d^t}{interval_i} \cdot \left[\sum_{j \in J} \left(\sum_{l \in J} GR_{i,k}^l \cdot \sigma^{l,j} \right) \cdot labor_rate_j + other_costs_i \right] \quad (\text{B.8})$$

831 Constraint (B.2) guarantees that each task item is allocated exactly once. Con-
 832 straints (B.3) ensures that the available labor hours for each skill type is not
 833 exceeded in each of the maintenance time segments. The other three set of
 834 constraints (B.4)–(B.6) are the maintenance time-intervals constraint.

835 We developed a task allocation heuristic based on the worst-fit decreasing
 836 algorithm (WFD). First of all, we divide the entire maintenance check schedule
 837 into several time segments, according to the periods of overlapped maintenance
 838 checks (i.e., periods that several aircraft having the same type of maintenance
 839 checks). Next, we sort the time segments according to their associated available
 840 labor-hours in descending order. After that, we allocate each maintenance task
 841 to the last possible segment before its due dates. The detail of the task allocation
 842 can be found in [30].

843 *Nomenclature*

844 **Sets**

845 i : task indicator
 846 K : set of aircraft
 847 N_k : set of task items for aircraft k ($k \in K$)
 848 T_k : set of time segments for aircraft k ($k \in K$)
 849 $R_{i,k}$: set of time segments for task item i ($i \in N_k$) of aircraft k ($k \in K$)
 850 J : set of skills
 851 $O_{i,k}$ unit set with the task item that follows task item i ($i \in N_k$) of aircraft
 852 k ($k \in K$)

853 **Parameters**

854 $c_{i,k}^t$: cost of allocating task item i ($i \in N_k$) from aircraft k ($k \in K$) to
 855 maintenance opportunity belonging to time segment t ($t \in T_k$)
 856 $\overline{GR}_t^j$: amount of available labor hours of skill type j ($j \in J$) at time
 857 segment t
 858 $GR_{i,k}^j$: amount of labor hours of skill type j prescribed to perform task
 859 item i of aircraft k

$\sigma^{j,l}$: “non-routine rate” indicating the amount of labor hours needed from
skill type l for every labor-hour prescribed from skill type j (note: $\sigma^{j,j} \geq$
 $1.0 \quad \forall j \in J$)
 $\bar{d}_{i,k}$: maximum number of days between rescheduling task item i ($t \in T_k$)
for aircraft k ($k \in K$)
 d^t : number of days from the start of the planning horizon till maintenance
opportunity belonging to time segment t
 $\bar{fh}_{i,k}$: maximum number of flight-hours between rescheduling task item i
for aircraft k
 fh^t : number of accumulated flight-hours from the start of the planning
horizon till maintenance opportunity belonging to time segment t
 $\bar{fc}_{i,k}$: maximum number of flight-cycles between rescheduling task item i
for aircraft k
 fc^t : number of accumulated flight-cycles from the start of the planning
horizon till maintenance opportunity belonging to time segment t
 O_{day_i} : total days of aircraft operations from the start of the planning
horizon to the due date of performing task item i , following the task fix
interval and if no resource constraints are considered
 $interval_i$: average fix interval for task item i measured in days
 $labor_rate_j$: labor rate, per hour, of skill type j ($j \in J$)
 $other_costs_{i,k}$: non-labor costs associated with task item i ($i \in N_k$) of
aircraft k ($k \in K$), such as costs of spare parts and tooling

Decision variables

$x_{i,k}^t$: 1 if task item i is assigned to maintenance opportunity belonging to
time segment t for aircraft k , and 0 otherwise